

In practice, the L_0 loss on the last term is relaxed to L_p for $0 < p \leq 1$ to make the loss differentiable. We call the m columns of $\mathbf{D}$ (vectors in $\mathbb{R}^d$) **dictionary elements**.

We now argue that SAEs can discover irreducible multi-dimensional features by *clustering* $\mathbf{D}$. We will consider a simple form of clustering: build a complete graph on $\mathbf{D}$ with edge weights equal to the cosine similarity between dictionary elements, prune all edges below a threshold T , and then set the clusters equal to the connected components of the graph. If we now consider the spaces spanned by each cluster, they will be approximately T -orthogonal by construction, since their basis vectors are all T -orthogonal. Now, consider some irreducible two-dimensional feature $\mathbf{f}$; we claim that if the SAE is large enough and $\mathbf{f}$ is active enough such that the SAE can reconstruct $\mathbf{f}$ when $\mathbf{f}$ is active, one of the clusters is likely to be exactly equal to $\mathbf{f}$. If $\mathbf{D}$ includes just two dictionary elements spanning $\mathbf{f}$, then these elements both must have nonzero activations post-ReLU to reconstruct $\mathbf{f}$ (otherwise $\mathbf{f}$ is a mixture). Because of the sparsity penalty in Eq. (4), this two-vector solution to reconstruct $\mathbf{f}$ is disincentivized, so instead the dictionary is likely to learn many elements that span $\mathbf{f}$. These dictionary elements will then have a high cosine similarity, and so the edges between them will not be pruned away during the clustering process; hence, they will be in a cluster.

Thus, we have a way to operationalize Hypothesis 2: clustering $\mathbf{D}$ finds T -orthogonal subspaces, and if irreducible multi-dimensional features exist, they are likely to be equal to some of these subspaces. This suggests a natural approach to using sparse autoencoders to search for irreducible multi-dimensional features:

1. Cluster dictionary elements by their pairwise cosine similarity. We use both the simple similarity-based pruning technique described above, as well as spectral clustering; see Appendix F for details, including comments on scalability.
2. For each cluster, run the SAEs on all $\mathbf{x}_{i,l} \in X_{i,l}$ and ablate all dictionary elements not in the cluster. This will give the reconstruction of each $\mathbf{x}_{i,l}$ restricted to the cluster found in step 1 (if no cluster dictionary elements are non-zero for a given point, we ignore the point).
3. Examine the resulting reconstructed activation vectors for irreducible multi-dimensional features. This step can be done manually by visually inspecting the PCA projections for known irreducible multi-dimensional structures (e.g. circles, see Fig. 10) or automatically by passing the PCA projections to the tests for Definition 3.

Pseudocode for this method is in the appendix in Alg. 1. This method succeeds on toy datasets of synthetic irreducible multi-dimensional features; see Appendix D.³ We apply this method to language models using GPT-2 (Radford et al., 2019) SAEs trained by Bloom (2024) for every layer and Mistral 7B (Jiang et al., 2023) SAEs that we train on layers 8, 16, and 24 (training details in Appendix E).

Strikingly, we reconstruct irreducible multi-dimensional features that are *interpretable* circles: in GPT-2, days, months, and years are arranged circularly in order (see Fig. 1); in Mistral 7B, days and months are arranged circularly in order (see Fig. 15). These plots contain the PCA dimensions that most clearly show circular structure; these best dimensions are usually the second and third because the first PCA dim is an “intensity” direction that manifests as the radius of the circle in Fig. 1 (thus the overall structure for these multi-d features is perhaps best thought of as a cone). See Fig. 14 for all PCA dimensions visualized).

For each cluster of GPT-2 SAE features, we take the reconstructed activations and project them onto PCA components 1-2, 2-3, 3-4, and 4-5 (or fewer if there are fewer features in the cluster) and measure the separability index and ϵ -mixture index of each 2D point cloud as described in Appendix B.2. The mean scores across these planes are a computationally tractable approximation of Definition 3. We plot these mean scores in Fig. 3, and find that the features which we had manually identified in Fig. 1 are among the top scoring features along

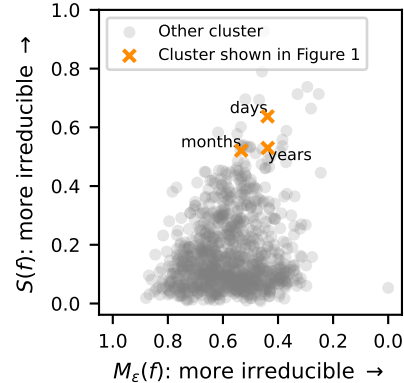


Figure 3: Mixture index and separability index of GPT-2 features. Features from Fig. 1, which we had manually identified, score highly as candidate multi-dimensional features with these metrics.

³Code: <https://github.com/JoshEngels/MultiDimensionalFeatures>

Table 1: Aggregate model accuracy on days of the week and months of the year modular arithmetic tasks. Performance broken down by problem instance in Appendix I.

Model	Weekdays	Months
Llama 3 8B	29 / 49	143 / 144
Mistral 7B	31 / 49	125 / 144
GPT-2	8 / 49	10 / 144

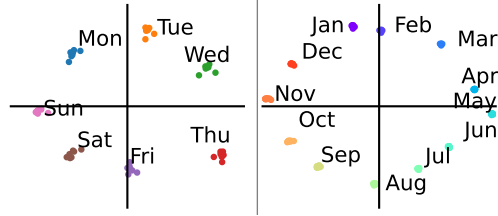


Figure 4: Top two PCA components on the α token. Colors show α . **Left:** Layer 30 of Mistral on Weekdays. **Right:** Layer 5 of Llama on Months.

both measures of irreducibility. Thus, our theoretical tests can indeed be used to find interpretable irreducible features. We show the top 20 feature clusters, measured by the product of $(1 - \epsilon\text{-mixture index})$ and separability index, in Appendix G. Out of all 1000 clusters, the Fig. 1 clusters rank 9, 28, and 15 by this metric, respectively.⁴

5 CIRCULAR REPRESENTATIONS IN LARGE LANGUAGE MODELS

In this section, we examine tasks in which models *use* the multi-dimensional features we discovered in Section 4, thereby providing evidence that these representations are indeed the fundamental unit of computation for some problems. Inspired by prior work studying circular representations in modular arithmetic (Liu et al., 2022), we define two prompts that represent “natural” modular arithmetic tasks:

Weekdays task: “Let’s do some day of the week math. Two days from Monday is”
Months task: “Let’s do some calendar math. Four months from January is”

For Weekdays, we range over the 7 days of the week and durations between 1 and 7 days to get 49 prompts. For Months, we range over the 12 months of the year and durations between 1 and 12 months to get 144 prompts. Mistral 7B and Llama 3 8B (AI@Meta, 2024) achieve reasonable performance on the Weekdays task and excellent performance on the Months task (measured by comparing the highest logit valid token against the ground truth answer), as summarized in Table 1. Interestingly, although these problems are equivalent to modular arithmetic problems $\alpha + \beta \equiv ? \pmod{m}$ for $m = 7, 12$, both models get trivial accuracy on plain modular addition prompts, e.g. “ $5 + 3 \pmod{7} \equiv$ ”. Finally, although GPT-2 has circular representations, it gets trivial accuracy on Weekdays and Months.

To simplify discussion, let α be the day of the week or month of the year token (e.g. “Monday” or “April”), β be the duration token (e.g. “four” or “eleven”), and γ be the target ground truth token the model should predict, such that (abusing notation) we have $\alpha + \beta = \gamma$. Let the prompts of the task be parameterized by j , such that the j th prompt asks about α_j , β_j , and γ_j .

We confirm that Llama 3 8B and Mistral 7B have circular representations of α on this task by examining the PCA projections of hidden states across prompts at various layers on the α token. We plot two of these in Fig. 4 and show all layers in Fig. 18. These plots show circular representations as the highest varying two components in the model’s representation of α at many layers.

5.1 INTERVENING ON CIRCULAR DAY AND MONTH REPRESENTATIONS

We now experiment with *intervening* on these circular representations. We base our experiments on the common interpretability technique of activation patching, which replaces activations from a “dirty” run of the model with the corresponding activations from a “clean” run (Zhang & Nanda, 2023). Activation patching empirically tests whether a specific model component, position, and/or representation has a causal influence on the model’s output. We employ a custom subspace patching

⁴We also tried an alternative ranking scheme: we sorted the clusters by separability and irreducibility and set the cluster score equal to the minimum sorted position between the two sorted lists. The Fig. 1 clusters rank 8, 105, and 12 by this metric.

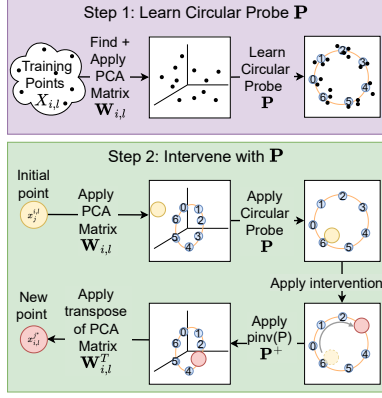


Figure 5: Visual representation of the circular intervention process. **Top:** We learn a circular probe on the PCA projection of a training set. **Bot:** To intervene, we change the circular representation to α'_j and average ablate other dimensions.

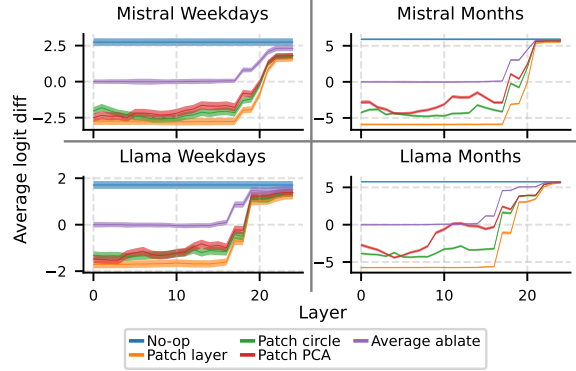


Figure 6: Mean and 96% error bars for intervening on the α token across layers using different intervention methods. The circular intervention technique outperforms patching only the top 5 PCA components and leaving the rest unchanged, and almost reaches the upper bound performance of patching the entire layer.

method to allow testing for whether a specific *circular subspace* of a hidden state is sufficient to causally explain model output. Specifically, our patching technique relies on the following steps (visualized in Fig. 5):

1. **Find a subspace with a circle to intervene on:** Using a PCA reduced activation subspace to avoid overfitting, we train a “circular probe” to identify representations which exhibit strong circular patterns. More formally, let $\mathbf{x}_{i,l}^j$ be the hidden state at layer l token position i for prompt j . Let $\mathbf{W}_{i,l} \in \mathbb{R}^{k \times d}$ be the matrix consisting of the top k principal component directions of $\mathbf{x}_{i,l}^j$. In our experiments, we set $k = 5$. We learn a linear probe $\mathbf{P} \in \mathbb{R}^{2,k}$ from $\mathbf{W}_{i,l} \cdot \mathbf{x}_{i,l}^j$ to a unit circle in α . In other words, if $\text{circle}(\alpha) = [\cos(2\pi\alpha/7), \sin(2\pi\alpha/7)]$ for Weekdays and $\text{circle}(\alpha) = [\cos(2\pi\alpha/12), \sin(2\pi\alpha/12)]$ for Months, $\mathbf{P}$ is defined as follows:

$$\mathbf{P} = \arg \min_{\mathbf{P}' \in \mathbb{R}^{2,k}} \sum_{\mathbf{x}_{i,l}^j} \left\| \mathbf{P}' \cdot \mathbf{W}_{i,l} \cdot \mathbf{x}_{i,l}^j - \text{circle}(\alpha) \right\|_2^2 \quad (5)$$

2. **Intervene on the subspace:** Say our initial prompt had $\alpha = \alpha_j$ and we are intervening with $\alpha = \alpha_{j'}$. In this step, we replace the model’s projection on the subspace $\mathbf{P} \cdot \mathbf{W}_{i,l}$, which will be close to $\text{circle}(\alpha_j)$, with the “clean” point $\text{circle}(\alpha_{j'})$. Note that we do not use the hidden state $\mathbf{x}_{i,l}^{j'}$ from the “clean” run, only the “clean” label $\alpha_{j'}$. In practice, other subspaces of $\mathbf{x}_{i,l}^j$ may be used concurrently by the model in “backup” circuits (see e.g. Wang et al. (2022)) to compute the answer, so if we just intervene on the circular subspace the remaining components of the activation may interfere in downstream computations. Thus, to isolate the effect of our intervention, we set the average ablate the portion of the activation not in the intervened subspace. Letting $\overline{\mathbf{x}_{i,l}}$ be the average of $\mathbf{x}_{i,l}^j$ across all prompts indexed by j and $\mathbf{P}^+$ be the pseudoinverse of $\mathbf{P}$, we intervene via the formula

$$\mathbf{x}_{i,l}^{j*} = \overline{\mathbf{x}_{i,l}} + \mathbf{W}_{i,l}^T \mathbf{P}^+ (\text{circle}(\alpha_{j'}) - \overline{\mathbf{x}_{i,l}}) \quad (6)$$

We run our patching on all 49 Weekday problems and 144 Month problems and use as “clean” runs the 6 or 11 other possible values for β , resulting in a total of $49 * 6$ patching experiments for Weekdays and $144 * 11$ patching experiments for Months. We also run baselines where we (1) replace the entire subspace corresponding to the first 5 PCA dimensions with the corresponding subspace from the clean run, (2) replace the entire layer with the corresponding layer from the clean run, and (3) replace the entire layer with the average across the task. The metric we use is *average logit difference* across all patching experiments between the original correct token (α_j) and the

target token ($\alpha_{j'}$). See Fig. 6 for these interventions on all layers of Mistral 7B and Llama 3 8B on Weekdays and Months.

The main takeaway from Fig. 6 is that circular subspaces are causally implicated in computing γ , especially for Weekdays. Across all models and tasks, early layer interventions on the circular subspace have almost the same intervention effect as patching the entire layer, and are usually better than patching the top PCA dimensions from the clean problem.

Patching experiments in Appendix J show α is copied to the final token on layers 15 to 17, which is why interventions drop off there. Additionally, while in this section we train a probe on a dataset of prompts, in Section 5.2, we show that intervening on the circle discovered via SAE clustering in Section 4 also works.

To investigate exactly how models use the circular subspace, we perform *off distribution* interventions. We modify Eq. (6) so that instead of intervening on the circumference $\text{circle}(\alpha)$, we sweep over a grid of positions (r, θ) within the circle:

$$\mathbf{x}_{i,l}^{j*} = \overline{\mathbf{x}_{i,l}} + \mathbf{W}_{i,l}^T \mathbf{P}^+ [r \cos(\theta), r \sin(\theta)]^T - \overline{\mathbf{x}_{i,l}} \quad (7)$$

We intervene with $r \in [0, 0.1, \dots, 2], \theta \in [0, 2\pi/100, \dots, 198\pi/100]$ and record the highest logit γ after the forward pass. Fig. 7 displays these results on Mistral layer 5 for $\beta \in [2, 3, 45]$. They imply that Mistral treats the circle as a multi-dimensional representation with α encoded in the angle.

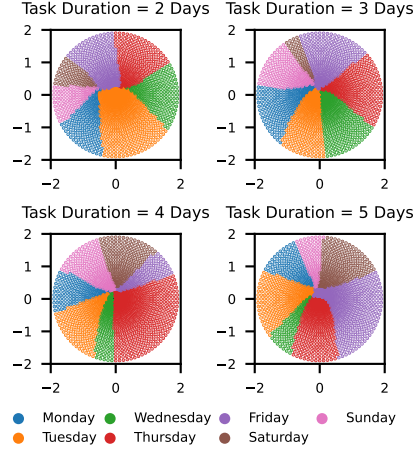


Figure 7: Off distribution interventions on Mistral layer 5 on the Weekdays task. The color corresponds to the highest logit γ after performing the circular subspace intervention on that point.

5.2 INTERVENING WITH THE SAE PLANE

In the last section, we train probes manually on the PCA of the activations to fit a circle. A perhaps more natural approach is intervening with the precise circle we found in Section 4. To determine if this approach is feasible, we first project layer 8 Mistral 7B Weekdays activations into the weekdays plane that was discovered by clustering (see Fig. 15; the plane is defined by PCA dimensions 2 and 3 of the cluster). In the rest of this section, we call this plane the SAE plane. In Fig. 19, we find that indeed, the Weekdays representations projected into the SAE plane form a circle (see Fig. 19). We thus can fit a circular probe to this 2D plane as in Eq. (5). Similarly, we call this probe the SAE probe.

Because we only have layer 8 clustering results for Mistral, we train an SAE probe only on layer 8. We then evaluate interventions using this SAE probe on layer 8 of Mistral, but also at neighboring layers, since nearby layers should have similar representations (see e.g. Belrose et al. (2023)). We compare to two baselines: 1) training a normal circular probe on the PCA projections of each layer as described in Section 5.1, and 2) training a circular probe only on layer 8 and then evaluating on adjacent layers (in the same way as for the layer 8 SAE probe). We show the results of these methods in Fig. 8.

We find that on all layers, using the SAE probe only slightly decreases intervention performance as compared to training a circular probe (from -2.58 to -2.01 average logit difference on layer 8). Even more interestingly, the layer 8 SAE probe is much more robust to layer shifts than the layer 8 circular probe; for example, using the layer 8 circular probe on layer 6 results in an average logit difference of 0.029, whereas using the layer 8 SAE probe results in an average logit difference of

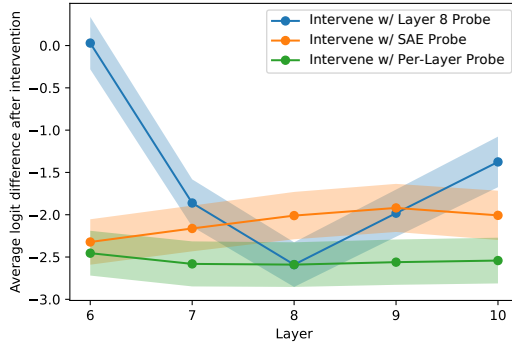


Figure 8: Interventions on the Mistral 7B Weekdays task with different methods of determining the probe.

-2.32. This is intriguing evidence that the SAE is perhaps finding more “true” (or at least more robust) features than our circular probing technique.

5.3 CONTINUITY OF CIRCULAR REPRESENTATIONS

In past sections, the representations of the interpretable numeric quantities we have discovered have been mostly *discontinuous*; that is, the days of the week and months of the year in Fig. 1 and Fig. 15 are clustered at the vertices of a heptagon and dodecagon, and there is nothing “between” adjacent weekdays or months along the circle. In this section, we will examine the “continuity” of the circular features we have discovered. Although continuity of the representation is not a requirement of Definition 3, it would further decrease the ϵ -mixture index, and would also increase our subjective perception of the circular feature as an intrinsic model feature representing a continuous quantity (time). Thus, we create a synthetic dataset containing the text “[very early/very late] on [Monday/Tuesday/.../Sunday]” and simply plot the projections of the layer 30 activations into the top two PCA components of the activations of [Monday/Tuesday/.../Sunday].

The results, shown in Fig. 9, show that Mistral 7B indeed can map intermediate quantities to their expected place in the circle: the very early and very late version of each weekday are more towards the last and the next weekday along the circle, respectively. We show similar results for “[morning/evening] on [Monday/Tuesday/.../Sunday]” in Appendix Fig. 22.

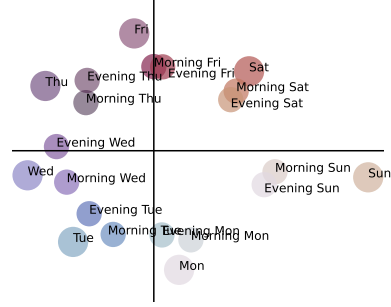


Figure 9: Layer 30 Mistral 7B activations for [morning/evening] on [Monday/Tuesday/.../Sunday], plotted projected into the PCA plane for [Monday/Tuesday/.../Sunday].

6 DISCUSSION

Our work proposes a significant refinement to the simple one-dimensional linear representation hypothesis. While previous work has convincingly shown the existence of one-dimensional features, we find evidence for irreducible multi-dimensional representations, requiring us to generalize the notion of a feature to higher dimensions. Fortunately, we find that existing unsupervised feature extraction methodologies like sparse autoencoders can readily be applied to discover multi-dimensional representations. However, we think our work raises interesting questions about whether individual SAE features are appropriate “mediators” (Mueller et al., 2024) for understanding model computation, if some features are in fact multi-dimensional. Although taking a multi-dimensional representation perspective may be more complicated, we believe that uncovering the true (perhaps multi-dimensional) nature of model representations is necessary for discovering the underlying algorithms that use these representations. Ultimately, our field aims to turn complex circuits in future more-capable models into formally verifiable programs (Tegmark & Omohundro, 2023; Dalrymple et al., 2024), which requires the ground truth “variables” of language models; we believe this work takes an important step towards discovering these variables.

Limitations: It is unclear why we did not find more interpretable multi-dimensional features. We are unsure if we are failing to interpret some of the high-scoring multi-dimensional features, if most multi-dimensional features lie in dimensions higher than two, if our clustering technique is not powerful enough to find some features, or if there are truly not that many. Additionally, our definitions for irreducible features (Definition 2) are purely statistical and not intervention based, and also had to be relaxed to hold in practice, resulting in measures that return a possibly subjective “degree” of reducibility (Definition 3). Thus, although this work provides preliminary evidence for the multi-dimensional superposition hypothesis (Hypothesis 2), it is still unclear if this theory provides the best description for the representations models use. Future work might make progress on this question by investigating new techniques for decomposing model representations, exploring higher dimensional representations, or determining conclusively whether models use representations in ways that necessitate the representations are non-linear.